1 a) $\dfrac{\partial F_1}{\partial y} = e^{y-x} + (y+\alpha)e^{y-x}$

$\dfrac{\partial F_2}{\partial x} = y e^{y-x}$

$\left.\begin{matrix} \dfrac{\partial F_1}{\partial y} = e^{y-x} + (y+\alpha)e^{y-x} \\ \dfrac{\partial F_2}{\partial x} = y e^{y-x} \end{matrix}\right\} \dfrac{\partial F_1}{\partial y} = \dfrac{\partial F_2}{\partial x} \iff$

$\iff (1+\alpha)e^{y-x} = 0 \iff \alpha = -1$

b) $F(x,y) = \left((y-1)e^{y-x}, -ye^{y-x} + \cos y\right)$

$\dfrac{\partial f}{\partial x} = (y-1)e^{y-x}$ (1) $f(x,y) = (1-y)e^{y-x}$

$\dfrac{\partial f}{\partial y} = -ye^{y-x} + \cos y$ (2)

Z (1): $f(x,y) = (1-y)e^{y-x} + C(y) \overset{(2)}{\Longrightarrow}$

$\Longrightarrow \cancel{-e^{y-x}} + (\cancel{1}-\cancel{y})e^{y-x} + C'(y) = \cancel{-y}e^{y-x} + \cos y$

$C'(y) = \cos y$

$C(y) = \sin y + K, \quad K \in \mathbb{R}$

$f(0,0) = 0 \Rightarrow 0 = 1 + C(0) = 1 + K \Rightarrow K = -1.$

Tedy

$f(x,y) = (1-y)e^{y-x} + \sin y - 1.$

② a) $\nabla f(x,y)=0 \ldots \quad 3x^2+2xy=0$
$\qquad\qquad\qquad\qquad x^2+2y-4=0$

$x(3x+2y)=0 \quad (1) \iff x=0$ nebo $3x=-2y$
$x^2+2y=4 \quad (2)$

- $x=0 \overset{(2)}{\Longrightarrow} y=2$
- $y=-\frac{3}{2}x \overset{(2)}{\Longrightarrow} x^2-3x-4=0$
  $(x-4)(x+1)=0$ — $x=4 \Rightarrow y=-6$; $x=-1 \Rightarrow y=\frac{3}{2}$

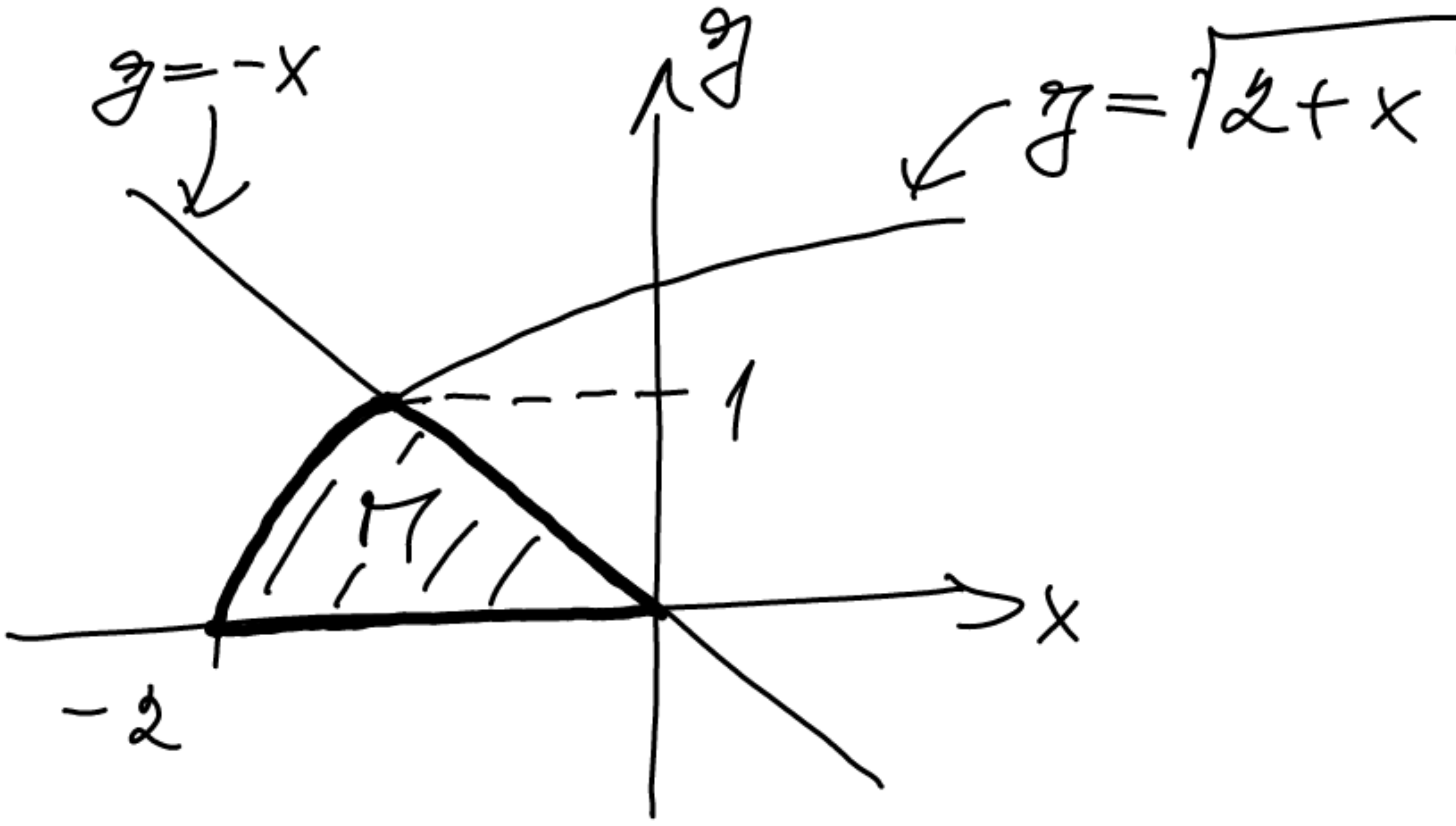
Stac. body funkce f: $(0,2), (4,-6), (-1,\frac{3}{2})$

$$H_f(x,y)=\begin{pmatrix} 6x+2y & 2x \\ 2x & 2 \end{pmatrix}$$

- $H_f(0,2)=\begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix} \ldots H_f(0,2)$ je pozit. def.
  $\Rightarrow (0,2)$ je bod lok. min.
- $H_f(4,-6)=\begin{pmatrix} 12 & 8 \\ 8 & 2 \end{pmatrix} \ldots \det H_f(4,-6)=24-64<0 \Rightarrow$
  $\Rightarrow (4,-6)$ je sedlový bod
- $H_f(-1,\frac{3}{2})=\begin{pmatrix} -3 & -2 \\ -2 & 2 \end{pmatrix} \ldots \det H_f(-1,\frac{3}{2})=-10<0 \Rightarrow$
  $\Rightarrow (-1,\frac{3}{2})$ je sedlový bod.

b) f není konvexní, protože $H_f(x,y)$ není pozitivně definitní na $\mathbb{R}^2$.

③



, dh: $x=y^2-2$

$$\int_M 1-y \, d\lambda_2 = \int_0^1 \int_{y^2-2}^{-y} 1-y \, dx \, dy$$

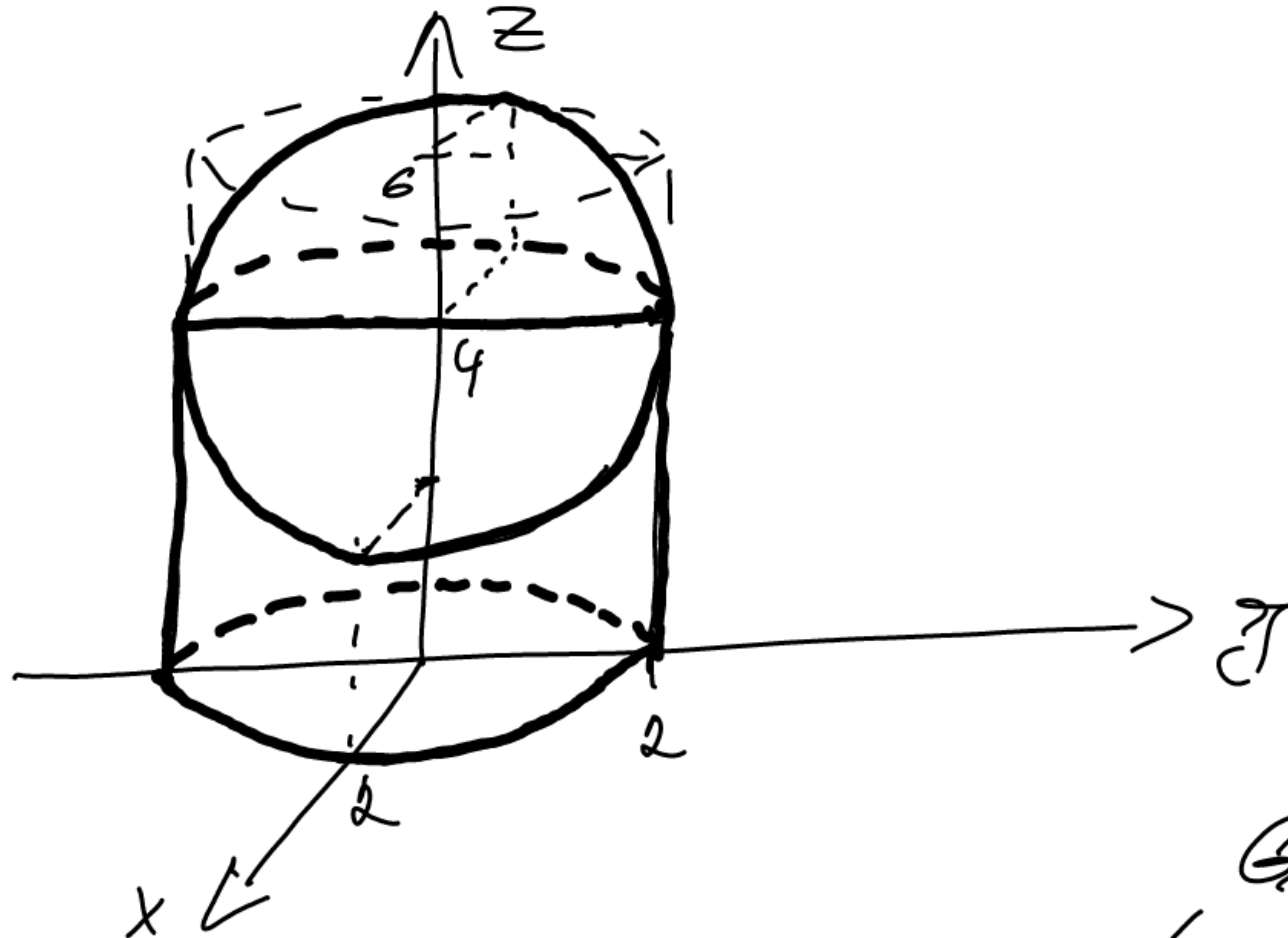
$$= \int_0^1 (1-y)(-y-y^2+2) \, dy$$

$$= \int_0^1 -y-\cancel{y^2}+2+\cancel{y^2}+y^3-2y \, dy = \int_0^1 y^3-3y+2 \, dy$$

$$= \left[\frac{y^4}{4} - \frac{3}{2}y^2 + 2y \right]_0^1 = \frac{1}{4} - \frac{3}{2} + 2$$

$$= \frac{1-6+8}{4} = \underline{\underline{\frac{3}{4}}}$$

4



Gaussova veta

$$\int_{\partial M} F \cdot d\sigma \overset{\text{Vále. souř.}}{=} \int_M \nabla \cdot F \, d\lambda_3 = \int_M 3x^2 + 3y^2 + x \, d\lambda_3$$

$$= \int_0^{2\pi}\int_0^{2}\int_0^{4-r\cos\varphi} \left(3r^2 + r\cos\varphi\right) r \, dz \, dr \, d\varphi$$

$$= \int_0^{2\pi}\int_0^{2} \left(3r^3 + r^2\cos\varphi\right)\left(4 - r\cos\varphi\right) dr \, d\varphi$$

$$= \int_0^{2\pi}\int_0^{2} 12r^3 + 4r^2\cos\varphi - 3r^4\cos\varphi - r^3\cos^2\varphi \, d\varphi$$

$$= \int_0^{2\pi} 48 + \left(\frac{4}{3}\cdot 8 - \frac{3}{5}\cdot 32\right)\cos\varphi - 4\,\frac{1+\cos(2\varphi)}{2} \, d\varphi = \underline{\underline{32\pi}}$$

5 $$\lim_{k\to\infty}\left|\frac{\frac{2k+3}{(-4)^{k+1}(k+1)!}\,x^{2k+2}}{\frac{2k+1}{(-4)^{k}k!}\,x^{2k}}\right|$$

$$=\lim_{k\to\infty}\frac{2k+3}{2k+1}\,\frac{1}{4}\,\frac{1}{k+1}\,|x|^2=0\;\Rightarrow\;R=+\infty$$

Pro $x\in(-\infty,\infty)$ máme

$$\sum_{k=0}^{\infty}\frac{2k+1}{(-4)^k k!}\,x^{2k}=\left(\sum_{k=0}^{\infty}\frac{x^{2k+1}}{(-4)^k k!}\right)'$$

$$=\left(x\sum_{k=0}^{\infty}\frac{\left(\frac{x^2}{-4}\right)^k}{k!}\right)'=\left(x e^{-\frac{x^2}{4}}\right)'$$

$$=e^{-\frac{x^2}{4}}-\frac{x^2}{2}e^{-\frac{x^2}{4}}=\left(1-\frac{x^2}{2}\right)e^{-\frac{x^2}{4}}$$

Mocninná řada konverguje v bodě 5, neboť interval konvergence je $(-\infty,\infty)$.